PROJECT PROPOSAL

Cyber-Physical Smart Building Access Control System

Revised Proposal — Technical Architecture, Security, Budget, and Delivery Plan

1. Introduction

Modern universities and large institutional buildings are moving toward automated, networked infrastructure. Manual door management — physical keys, unlogged access, no remote oversight — cannot meet the security and operational demands of a modern campus. Physical site visits to reset a stuck lock, manually admit a late instructor, or investigate an access dispute are slow, unauditable, and do not scale as a campus grows.

This project introduces a cyber-physical access control system that connects real-world door hardware to a secure digital management platform over the university network, replacing ad hoc manual processes with centralized, auditable, and remotely manageable control.

2. Project Overview

The Cyber-Physical Smart Building Access Control System ("the System") provides centralized control over building entrances and individual room access across a university campus. It bridges physical door hardware with a digital management layer accessible to authorized personnel through a secure web application.

The System supports three operational modes:

- Automated Scheduled Access — doors unlock and lock according to class timetable data synchronized from the registrar system.
- Manual Instructor/Admin Override — authorized personnel trigger door actions on demand, outside the scheduled timetable.
- Real-Time Synchronization & Logging — every access event, lock-state change, or unauthorized attempt is logged centrally as it happens.

3. System Architecture & Methodology

The System is organized into three intercommunicating layers, mirroring the edge / network / application structure used in industrial cyber-physical systems.

A. Edge Layer (Door Node)

Each controlled door has a node built around an ESP32 microcontroller. Entry is authenticated using a dual-factor credential: a MIFARE DESFire RFID/NFC card (chosen over legacy 125 kHz or MIFARE Classic cards, which are trivially cloned with inexpensive tools) combined with a fingerprint check for higher-security rooms. A magnetic reed switch independently reports whether the door is physically open or closed, decoupled from the commanded lock state, so the System can detect forced or propped-open doors rather than trusting the lock's own status.

Every door also carries a request-to-exit (REX) button and a mechanical key override, regardless of its configured lock behavior. This is a life-safety requirement, not an optional extra — it guarantees occupants always have a manual, non-networked way out.

B. Network & Communication Layer

The System uses a hybrid network topology: door nodes connect primarily over a wired RS-485 backbone to a local building controller, with each node's onboard Wi-Fi acting as an automatic fallback if the wired bus fails. This mirrors a failover principle common in industrial monitoring systems, applied here to a life-safety-adjacent system where communication uptime affects both security and code compliance. Credential and event data is transmitted using MQTT over TLS, keeping the link encrypted end to end while minimizing bandwidth.

C. Application & Management Layer

A backend service authenticates dashboard users, manages the credential database, ingests door events, and synchronizes classroom schedules from the registrar system (or a realistic mock of that data source, scoped to what is feasible without production access to university systems). A web dashboard gives administrators real-time door status, manual override controls, historical access logs, and alerts for anomalies such as forced entry or repeated failed authentication attempts.

4. Hardware Components

Component	Role	Example Part
ESP32 Development Board	Door node MCU; onboard Wi-Fi provides network fallback	ESP32-WROOM-32
RFID/NFC Reader	Primary credential; DESFire support resists cloning	PN532 module
Fingerprint Module	Second-factor credential for higher-security rooms	R307 / AS608
Door Position Sensor	Independent physical open/closed detection	Magnetic reed switch
Lock Actuator + Driver	Electric strike, opto-isolated relay/MOSFET driver	12V DC electric strike + relay module
RS-485 Transceiver	Wired backbone communication to building controller	MAX485 breakout
Request-to-Exit Button	Life-safety egress override, present at every door	Standard REX push button
Mechanical Key Override	Manual emergency/authorized bypass	Keyed bypass cylinder
Power Supply	Node power; brief local backup so logging survives short outages	12V PSU + small battery backup

5. Software & Technology Stack

Category	Tools / Technologies
Firmware	C/C++ via PlatformIO or ESP-IDF
Communication Protocol	MQTT over TLS (Mosquitto/HiveMQ broker); RS-485 serial driver on the wired backbone
Backend	Node.js/Express or Python/FastAPI; JWT-based authentication
Database	PostgreSQL — users, credentials, schedules, audit logs
Dashboard	React (custom build) or Node-RED (faster to build, less flexible)
Dev & Test Tools	VS Code, Wireshark (protocol/TLS verification), Postman (API testing)

6. Security & Fail-Safe Design

Credential security: card credentials use MIFARE DESFire with AES-based mutual authentication rather than legacy cloneable cards. Fingerprint data is stored as encrypted templates, never raw images. All credential and event traffic is encrypted in transit (TLS) and at rest in the database.

Tamper detection: each node reports a case-open/tamper switch state to the backend; a node going offline or reporting tamper triggers an immediate dashboard alert.

Fail-safe / fail-secure policy: standard and high-security rooms default to fail-secure — doors remain locked on power loss, prioritizing security over convenience. Every fail-secure door is paired with a REX button and mechanical key override so this default never removes a manual way out. Doors on designated fire egress routes are configured fail-safe (unlock on power loss) regardless of the general default, as required by applicable fire and life-safety code. The System supports this as a per-door configuration rather than a single global setting, since a real building always needs both behaviors depending on the door's role.

7. Project Impact

- Improved Security — restricts building access to authenticated, dual-factor credentials and eliminates cloneable legacy cards.
- Operational Efficiency — removes manual keys and physical round-checks, reducing dedicated facilities staff time.
- Classroom Automation — bridges the registrar's scheduling data with real-world door state, so rooms are ready without manual intervention.
- Centralized Dashboard — a single environment for monitoring and control across multiple doors, floors, or buildings.
- Scalable Architecture — modular node design allows additional doors or buildings to be added without redesigning the System.

8. Related Work & Market Positioning

Commercial access control platforms — HID Global, Salto, Genetec, and similar — already serve this market, typically as proprietary, subscription-licensed systems with limited customization. This project's contribution is not a novel access-control concept, but a self-built, campus-specific implementation that integrates directly with the registrar's class schedule — a feature most commercial platforms do not offer without costly custom integration — while remaining fully open and modifiable by the institution that owns it.

Academically, the project demonstrates competency across embedded systems, secure networking, backend and database design, and interface design, aligned with current demand for engineers with cyber-physical systems experience in smart infrastructure and building automation.

9. Team Structure & Roles

Role	Responsibility
Embedded/Firmware Engineer	Door-node MCU code: credential reading, lock actuation, door-state sensing
Hardware/Circuit Engineer	Lock driver circuit, power supply and isolation, sensor wiring, schematic/PCB
Backend/Database Engineer	API server, authentication, scheduling engine, credential and audit-log storage
Frontend/Dashboard Developer	Admin web application: door status, overrides, logs, alerts
Network/Security Engineer	Protocol selection, encryption, tamper detection, fail-safe logic, integration testing
QA/Documentation (6th member, if available)	Test plan execution, final report assembly, evaluation results

10. Project Timeline

The project is planned over a 34-week schedule, tracked in detail in an accompanying task-level Gantt tracker. Phase summary:

Phase	Weeks
0 — Proposal & Planning	1–3
1 — System Design	3–6
2 — Single Door-Node Prototype	6–11
3 — Backend & Database	8–14
4 — Dashboard	11–17
5 — Security & Fail-Safe Implementation	15–19
6 — Multi-Node Scaling	18–23
7 — Full System Testing	23–27
8 — Documentation	26–31
9 — Presentation & Defense Prep	31–34

11. Budget & Bill of Materials

Estimated cost for a 5-door pilot deployment, based on typical hobbyist/prototyping-grade component pricing. These are indicative planning figures, not supplier quotes — confirm current prices before purchasing.

Item	Unit Cost (est.)	Qty	Subtotal
ESP32 development board	$12	5	$60
PN532 NFC/RFID reader	$12	5	$60
R307 fingerprint module	$12	5	$60
Magnetic reed door sensor	$3	5	$15
Electric strike lock (12V, fail-secure)	$25	5	$125
Relay/MOSFET driver module	$3	5	$15
MAX485 RS-485 transceiver	$2	5	$10
REX button + mechanical key override	$13	5	$65
Enclosure, wiring, misc. per node	$15	5	$75
Central RS-485 gateway / controller	$20	1	$20
Backbone cabling	$50	1	$50
Contingency (≈ 15%)	—	—	$80

Estimated total: approximately $635 for a 5-door pilot. Scaling to a full building will scale roughly linearly per door plus a fixed backbone/controller cost.

12. Risk Assessment & Mitigation

Risk	Impact	Mitigation
Component/lead-time delays (strikes, biometric modules)	Schedule slip	Order all hardware in Phase 0, before design work starts; keep a spare of each critical part
Fail-secure default conflicts with fire/life-safety code on egress doors	Safety failure, project rejection	Per-door configurable fail-safe override + REX button and key bypass on every door, not just egress routes
Biometric false-reject/accept rate too high in practice	Unreliable demo	Card credential as primary, biometric as second factor only where required; test thoroughly in Phase 2
Campus network/IT restricts new devices	Integration blocked	Engage campus IT early about network access, VLANs, and firewall rules for the gateway and MQTT broker
RFID/NFC credential cloning	Security breach	MIFARE DESFire (AES mutual auth) instead of legacy cards; validate during security testing
Uneven team workload / member unavailability	Schedule slip	Cross-train on adjacent roles; track weekly progress against the Gantt tracker

13. Test & Evaluation Plan

- Door unlock latency (credential scan to unlock) — target under 1 second.
- Failover switch time from wired backbone to Wi-Fi fallback — measured and reported against a defined target.
- Biometric false accept/reject rate over a defined number of test attempts.
- Encryption verification — confirm TLS on MQTT traffic via packet capture; confirm no plaintext credential data anywhere in transit.
- Fail-safe/fail-secure validation — simulate power loss on both door types and confirm correct behavior.
- Dashboard load test — multiple door nodes reporting concurrently.
- Tamper detection test — trigger the case-open switch and confirm an alert is raised within a defined time window.

14. Conclusion

The Cyber-Physical Smart Building Access Control System replaces manual, unauditable door management with a centralized, dual-factor, fail-safe-aware control platform. This revision grounds the original concept in a concrete architecture, a specified hardware and software stack, an explicit security and life-safety design, and a realistic budget, timeline, and risk plan — the elements a technical committee will look for before approving fabrication and development work.